AC4409

Corporate Financing

Winter 2023 Examination

Paper 1 · 1.5 hours · Answer 2 of 3 questions · Show ALL workings

Module	AC4409 — Corporate Financing
Session	Winter 2023
Structure	Three questions, each worth 50 marks. Answer any two — all questions carry equal weight.
This scheme covers	All three questions in full — Q1, Q2, and Q3 — so you can study every possible pairing, not just the two you'd pick on the day.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the method marks for those later steps. This applies throughout the EOQ and leverage workings below.

Two source-paper notes. Question 1(b) states the cash discount is 1.5%, but also describes the revised terms as “3/30 net 60” — the 3 doesn't match the stated 1.5% and appears to be a typo; this scheme uses the explicitly stated 1.5% throughout. Question 3(b)'s introduction states MST generated €93,000 in revenue this year, which doesn't match 6,000 hips × €24 = €144,000 used in the actual calculations (i)–(vii); since those calculations are self-consistent and directly tied to the figures given for them, they're used as the basis throughout — see the checknote in Question 3(b) for the full reasoning.

Question 1

Inventory Management, Credit Policy & Working Capital (50 Marks)

Shoe Haven Ltd, a shoe shop in Cork City, operates with fixed ordering and holding costs. During a 180-day period, they sell 900 pairs of shoes. Ordering costs are €50 per order and holding costs are €15 per unit per 180 days, where a unit is a pair of shoes.

Part (a)(i)

2 Marks

(i) Calculate Shoe Haven Ltd's Economic Ordering Quantity (EOQ).

FULL MODEL ANSWER

EOQ = √(2DS ÷ H)
EOQ = √(2 × 900 × 50 ÷ 15)
EOQ = √6,000
EOQ ≈ 77.46 units (pairs of shoes)

Part (a)(ii)

2 Marks

(ii) Using the EOQ calculated in part (i), calculate how often Shoe Haven Ltd has to order the product in each 180-day period.

FULL MODEL ANSWER

Number of orders per period = D ÷ EOQ
= 900 ÷ 77.46
≈ 11.62 orders per 180-day period (roughly every 15.5 days)

Part (a)(iii)

6 Marks

(iii) Calculate Shoe Haven Ltd's holding costs, ordering costs and total inventory management costs, if it orders at the EOQ level of inventory each time.

FULL MODEL ANSWER

Holding cost = (EOQ ÷ 2) × H = (77.46 ÷ 2) × €15	€580.95
Ordering cost = (D ÷ EOQ) × S = (900 ÷ 77.46) × €50	€580.95
Total inventory management cost	**€1,161.90**

At the EOQ, holding cost and ordering cost are always exactly equal — that's the defining property of the EOQ (it's the order quantity that minimises the sum of the two, which occurs precisely where they intersect). If your two figures don't come out equal, that's a strong signal to recheck the EOQ calculation in part (i) before proceeding.

Part (a)(iv)

5 Marks

(iv) Previously the supplier took four days to deliver each pair of shoes ordered. Shoe Haven Ltd have been informed that due to a courier change, the lead time on their stock purchases may vary between four and six days. Calculate the reorder level which should ensure that there are no stock-outs.

FULL MODEL ANSWER

To guarantee no stock-outs under a variable lead time, the reorder level must be set using the **maximum** possible lead time (6 days), not the average — otherwise a delivery arriving on the slow end of the range would leave the shop without stock.

Daily demand = 900 units ÷ 180 days
= 5 units per day

Reorder level = Maximum lead time × daily demand
= 6 days × 5 units/day
= 30 units (pairs of shoes)

Equivalently: reorder at the average lead-time usage (4 days × 5 units = 20 units) **plus** a safety stock buffer covering the extra 2 days of potential delay (2 × 5 = 10 units), giving the same 30-unit answer. Both framings are acceptable, provided the reorder level ends up covering the worst case (6-day) lead time, not just the average.

How the 15 Marks for Part (a) Are Likely Allocated

Tier	Marks	What separates this tier
(i) 2	2	Correct EOQ formula and arithmetic, reaching ≈77.46 units.
(ii) 2	2	Correct division of demand by EOQ to reach ≈11.62 orders per period.
(iii) 6	6	Correct holding cost, ordering cost (both €580.95, confirming they're equal at EOQ), and total cost of €1,161.90; partial credit for a correct method with an EOQ figure carried over from an incorrect part (i).
(iv) 5	5	Correctly uses the MAXIMUM lead time (6 days, not the average or minimum) to calculate a reorder level of 30 units; marks lost for using average lead time without adding a safety stock buffer to cover the same worst case.

Part (b) **10 Marks**

EatEase plc currently offers 45 days credit to its customers (i.e. credit terms are net 45). Annual sales amount to €350,000 and are spread evenly over the year. The company's cost of funds is 6% per annum. A cash discount of 1.5% for payment by customers within 30 days is currently being considered (i.e. the revised credit terms would be 3/30 net 60). It is expected that 50% of customers would avail of the discount. Assuming that all sales are on credit and that there are 365 days per year, determine whether the discount should be offered.

You should round each calculation to the nearest euro. Show all your calculations and briefly explain your answer.

FULL MODEL ANSWER

The question states the discount is **1.5%** but then describes the new terms as "3/30 net 60" — the 3 doesn't match the 1.5% figure and appears to be a typo (should likely read "1.5/30 net 60"). The calculation below uses the explicitly stated 1.5% discount throughout, since that figure is unambiguous.

Step 1 — Average collection period, before and after

Under the new terms, the 50% of customers who take the discount pay in 30 days; the other 50% who don't take it pay at the new full net period of 60 days:

	Days
Old average collection period (all customers pay in 45 days)	45.00
New average collection period (50% × 30 + 50% × 60)	**45.00**

The average collection period comes out **identical** under both policies — purely a feature of these particular numbers (the 50/50 split happens to land exactly back on 45 days). This means the investment tied up in receivables doesn't change at all, which turns out to be the crux of the whole decision.

Step 2 — Investment in receivables, before and after

	€
Daily sales = €350,000 ÷ 365	959
Old investment in receivables = €959 × 45 days	43,151
New investment in receivables = €959 × 45 days	**43,151**

Change in investment in receivables = €43,151 − €43,151 = €0
Financing cost saving = 6% × €0 = €0

Step 3 — Cost of the discount

Cost of discount = Discount % × % of customers taking it × Annual sales
= 1.5% × 50% × €350,000
= €2,625

Step 4 — Decision

	€
Financing cost saving (benefit)	0
Less: cost of discount	(2,625)
Net effect of offering the discount	**(2,625)**

EatEase should NOT offer the discount. Because the average collection period happens to be unchanged, offering the discount frees up no cash at all — there's no financing-cost saving to offset the €2,625 the company gives away in discounts to the 50% of customers who take it. Offering the discount would be a straightforward net cost with no compensating benefit.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly calculates the (unchanged) average collection period under the new terms, correctly shows the investment in receivables is unchanged (so there's no financing saving), correctly calculates the €2,625 discount cost, and concludes the discount should not be offered with clear reasoning.
5–7	5–7	Correct method but an error in the average collection period calculation (e.g. assuming all customers now pay in 30 days, or omitting the 60-day payers), carried through under the own-figure rule.
2–4	2–4	Calculates the cost of the discount correctly but doesn't properly assess the financing benefit side, or reaches a conclusion without supporting calculations.
0–1	0–1	No attempt, or no structured comparison of costs and benefits.

Part (c) **25 Marks**

You are the financial analyst for a small manufacturing business. The company has been experiencing cash flow issues. Analyse the working capital management strategies you would recommend to address this challenge.

FULL MODEL ANSWER

A structured response works through each element of working capital in turn — inventory, receivables, payables, and cash itself — since a manufacturing business facing cash flow strain typically has cash trapped in more than one of these simultaneously.

Inventory management: applying EOQ discipline and reducing excess stock. A small manufacturer often carries more raw material and finished goods inventory than it needs, tying up cash that could otherwise ease the cash flow strain. Applying an EOQ-based ordering policy (as in part (a) above) rather than ad hoc ordering minimises the combined holding and ordering cost, and adopting just-in-time principles where the supply chain allows reduces the amount of capital sitting idle in the warehouse. A full inventory review to identify slow-moving or obsolete stock that can be discounted and cleared also converts genuinely stagnant working capital straight back into cash.

Receivables management: tightening credit policy without losing sales. Reviewing the credit terms currently extended to customers, tightening credit checks on new accounts, and actively chasing overdue balances can shorten the collection period and bring cash in faster. Where appropriate, a cash discount for early payment (weighed carefully using the cost/benefit framework applied in part (b) above — a discount is only worth offering if the financing benefit from a shorter collection period exceeds its cost) can accelerate collections. Invoice factoring or discounting is worth considering as a more immediate fix if receivables are large relative to the business's size, converting outstanding invoices into immediate cash at a cost.

Payables management: extending supplier terms without damaging relationships. Negotiating longer payment terms with key suppliers effectively uses supplier credit to fund the operating cycle rather than the company's own cash or a bank facility — but this has to be balanced against the risk of losing early-payment discounts already in place, and against straining supplier relationships if taken too far, which could jeopardise supply continuity precisely when the business can least afford disruption.

Cash management: forecasting and building a buffer. A rolling cash flow forecast (weekly or monthly, depending on the severity of the issue) gives early warning of upcoming shortfalls before they become a crisis, and lets the business plan financing needs proactively rather than reactively. Where a genuine short-term financing gap remains after tightening inventory, receivables and payables, an overdraft facility or short-term revolving credit line (as discussed in the broader financing-tools literature) can bridge the remaining gap while the underlying working capital improvements take effect.

Using the cash conversion cycle as the diagnostic and target metric. Calculating the cash conversion cycle (DSO + DIO − DPO) both now and as a target gives a single, trackable measure of whether the combined effect of these strategies is actually working, and highlights which of the three components (receivables, inventory, or payables) is the biggest drag on cash so effort can be prioritised there first, rather than spreading limited management attention evenly across all three regardless of where the real problem lies.

A strong answer doesn't treat these strategies as independent levers to pull in isolation — it recognises the trade-offs between them (e.g. tightening credit terms too aggressively risks losing sales to competitors offering easier terms; stretching payables too far risks supplier goodwill) and prioritises based on where the specific business's cash is most trapped, rather than offering a generic checklist of textbook working capital techniques.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
19–25	19–25	Covers all four working capital elements (inventory, receivables, payables, cash) with specific, actionable recommendations for each, explicitly weighs trade-offs between strategies, and ties the analysis together with a clear diagnostic approach (e.g. the cash conversion cycle) rather than a disconnected list.
13–18	13–18	Covers three of the four elements well, or all four with less depth and limited discussion of trade-offs.
7–12	7–12	General discussion of working capital management without specific, actionable recommendations tailored to a small manufacturing business.
0–6	0–6	Minimal or generic discussion with little real engagement with the scenario.

Question 1 Total: 50 Marks

Question 2

Financing Sources, WACC & Share Valuation (50 Marks)

Part (a) **15 Marks**

A company needs €100,000 for a six-month period to meet its financial requirements. Outline two sources of short-term funding that the company might consider. In your answer, you must outline the advantages and disadvantages of the sources selected.

FULL MODEL ANSWER

For a defined, relatively short (six-month) funding gap of €100,000, two well-suited sources are:

1. Bank Overdraft

Advantages: Highly flexible — the company only draws down (and pays interest on) what it actually needs at any given time, rather than the full €100,000 sitting as a fixed obligation throughout the six months; quick and straightforward to arrange with an existing bank relationship; and no long-term commitment beyond the facility's review period.

Disadvantages: Technically repayable on demand, so it isn't a fully reliable source over even a six-month horizon if the bank's own risk appetite changes; interest rates are typically higher than a committed term facility; and the bank may require security or impose covenants that constrain the business.

2. Short-Term Bank Loan

Advantages: A fixed-term loan matched to the six-month need gives certainty — the facility can't be withdrawn early the way an overdraft technically can — and a fixed repayment schedule makes cash flow planning straightforward over the period.

Disadvantages: Requires a credit application and, often, security or a personal guarantee for a smaller company; the full amount is typically drawn and incurring interest from day one, even if the company doesn't need the entire €100,000 immediately; and early repayment (if the need turns out to be shorter than expected) may attract a penalty.

Other reasonable choices include **trade credit** (if the €100,000 need relates to purchases from suppliers who'd extend it) or **invoice discounting** (if the company has receivables to draw against) — either would also be creditable if the advantages and disadvantages are correctly and specifically explained.

How the 15 Marks Are Likely Allocated

Tier	Marks	What separates this tier
12–15	12–15	Identifies two genuinely short-term financing sources appropriate to a six-month, €100,000 need, with a specific, well-explained advantage and disadvantage for each.
7–11	7–11	Two sources identified but with generic or thin advantages/disadvantages, or only one source fully developed.
3–6	3–6	Only one source properly identified and evaluated, or sources chosen that are not genuinely short-term in nature.
0–2	0–2	No attempt, or sources named with no advantages/disadvantages at all.

Part (b) **20 Marks**

A company needs €5 million in long-term funding to support its financial operations. Outline two sources of long-term funding that the company might consider. In your answer, you must outline the advantages and disadvantages of the sources selected.

FULL MODEL ANSWER

At €5 million and for the long term, the natural options move up a scale from the short-term tools in part (a):

1. Long-Term Bank Loan / Term Loan

Advantages: A term loan with a repayment schedule matched to the useful life of whatever the funding supports gives predictable, fixed interest costs (if fixed-rate) and preserves full ownership and control, since lenders have no equity stake or voting rights; interest is also typically tax-deductible, providing a tax shield the company wouldn't get from equity financing.

Disadvantages: Requires security (often a fixed or floating charge over company assets) and typically financial covenants that constrain the company's future decisions (e.g. restricting further borrowing or dividend payments); fixed repayment obligations regardless of how the business performs increase financial risk and the possibility of distress in a downturn; and a company already carrying significant debt may find €5 million difficult or expensive to raise this way.

2. Equity Issue (Share Capital)

Advantages: No fixed repayment obligation and no contractual interest cost — returns to shareholders (dividends) are entirely discretionary and can be reduced or skipped if the business underperforms, which reduces financial risk relative to debt; raising equity also increases the company's capacity to raise further debt in future, since it improves the balance sheet's gearing ratio.

Disadvantages: Dilutes existing shareholders' ownership percentage and voting control; new shares issued when the market believes the stock is fairly or richly valued can be read as a negative signal by the market (per Myers and Majluf's signalling theory), often depressing the share price; and equity is generally a more expensive source of capital than debt once the tax shield on interest is taken into account, since investors demand a higher return to compensate for bearing more risk than a lender.

Other strong alternatives include a **corporate bond issue** (suitable at this scale for a larger or credit-rated company, offering fixed long-term financing without the covenant intensity of a bank loan, but requiring the fixed costs of a public issue and ongoing disclosure) or **leasing** (if the €5 million relates to specific equipment or property, avoiding a large upfront capital outlay but typically costing more over the asset's full life than outright purchase) — either would also be creditable with properly explained advantages and disadvantages.

How the 20 Marks Are Likely Allocated

Tier	Marks	What separates this tier
15–20	15–20	Identifies two genuinely long-term financing sources appropriate to a €5 million need, with specific, well-explained advantages and disadvantages for each, ideally referencing the control/dilution and risk/flexibility trade-offs that distinguish debt from equity at this scale.
9–14	9–14	Two sources identified but with generic advantages/disadvantages, or only one source fully developed.
4–8	4–8	Only one source properly identified and evaluated, or sources chosen that aren't genuinely long-term in nature.
0–3	0–3	No attempt, or sources named with no advantages/disadvantages at all.

Part (c)

5 Marks

CelticCraft plc is considering a new investment project that is expected to generate cash flows for the next ten years. The company's target capital structure includes 60% equity and 40% debt. The cost of equity is estimated to be 12%, and the cost of debt is 5%. The corporate tax rate is 30%. Calculate the Weighted Average Cost of Capital (WACC) for CelticCraft.

FULL MODEL ANSWER

WACC = (E/V) × K_e + (D/V) × K_d × (1−T)
= (60% × 12%) + (40% × 5% × (1−30%))
= 7.20% + (40% × 5% × 70%)
= 7.20% + 1.40%
WACC = 8.60%

The 40% debt weight is multiplied by the **after-tax** cost of debt (5% × (1−30%) = 3.5%), not the raw 5% pre-tax figure — interest is tax-deductible, so the effective cost of debt to the company is lower than the contractual rate. Applying the tax adjustment to the equity component instead (or forgetting it altogether) is the most common error in this type of question.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Correctly weights each cost of capital by its target proportion and applies the tax shield to the cost of debt only, reaching 8.60%.
2–3	2–3	Correct formula but the tax adjustment omitted or misapplied.
0–1	0–1	No attempt, or the two costs simply averaged with no weighting.

Part (d) **10 Marks**

Answer either (d)(i) or (d)(ii).

Part (d)(i) **10 Marks**

Answer either (i): Explain how to value an ordinary share using the dividend discount model.

FULL MODEL ANSWER

The dividend discount model (DDM) values a share as the present value of all the future dividends an investor expects to receive from holding it, discounted at the investor's required rate of return on equity:

$$P_0 = \sum D_t \div (1+K_e)^t, \text{ for } t = 1 \text{ to } \infty$$

The general (multi-stage) form. In principle, every future dividend the company is expected to pay is forecast individually and discounted back to today at the investor's required return, then summed. In practice, forecasting an infinite stream of individual dividends isn't feasible, so the model is almost always simplified using an assumption about how dividends grow over time.

The zero-growth (perpetuity) special case. If dividends are assumed constant forever (no growth), the model collapses to a simple perpetuity: $P_0 = D \div K_e$, where D is the constant annual dividend.

The constant-growth (Gordon Growth) special case. If dividends are assumed to grow at a constant rate g forever, the model becomes $P_0 = D_1 \div (K_e - g)$, where D_1 is next year's expected dividend. This is the most commonly used version in practice, and is only valid when $K_e > g$ (otherwise the sum of discounted dividends never converges to a finite value).

The multi-stage (two- or three-stage) version. For a company expected to grow at a different (often higher) rate for a number of years before settling into a stable, mature growth rate, dividends are forecast explicitly year by year during the high-growth phase, discounted individually, and then a Gordon Growth terminal value is calculated at the point growth stabilises and discounted back to today alongside the explicit early dividends.

The DDM's main practical limitation is that it only works cleanly for companies that actually pay dividends, and requires an estimate of the growth rate g, which is often highly sensitive — a small change in the assumed g can swing the resulting valuation substantially, particularly as g approaches K_e.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly explains the general present-value-of-future-dividends concept AND at least the zero-growth and constant-growth special cases with their formulas, showing understanding of when each applies.
5–7	5–7	Explains the constant-growth (Gordon Growth) version correctly but with limited discussion of the general concept or other variants.
2–4	2–4	Vague description of "discounting dividends" without a specific formula or clear explanation of the growth assumption.
0–1	0–1	No attempt, or confuses the DDM with an unrelated valuation method.

Part (d)(ii) **10 Marks**

Or answer (ii): Aeroway Airlines plc has a current stock price of €50 and current dividend of €1.50. The dividend is expected to grow at 5% annually. Aeroway's beta is 0.85. The risk-free interest rate is 4.5%, and the equity risk premium is 6.0%. Value Aeroway using the Gordon growth model.

FULL MODEL ANSWER

The Gordon Growth Model needs a required rate of return — since one isn't given directly, it has to be derived from the beta, risk-free rate and equity risk premium using CAPM first.

Step 1 — Required return on equity (CAPM)

$K_e = R_f + \beta(\text{Equity Risk Premium})$
$= 4.5\% + 0.85 \times 6.0\%$
$= 4.5\% + 5.1\%$
$K_e = 9.6\%$

Step 2 — Next year's dividend

$D_1 = D_0 \times (1+g) = €1.50 \times 1.05$
$D_1 = €1.575$

Step 3 — Gordon Growth valuation

$P_0 = D_1 \div (K_e - g)$
$= €1.575 \div (9.6\% - 5\%)$
$= €1.575 \div 4.6\%$
$P_0 \approx €34.24$

The model's fair value of ≈€34.24 is notably **below** the given current market price of €50. This is worth stating explicitly in your answer: on the assumptions in the model (a 9.6% required return via CAPM and a constant 5% dividend growth rate), Aeroway's shares look **overvalued** in the market relative to what the Gordon Growth Model justifies. The €50 current price isn't an input the calculation itself needs — the Gordon Growth Model derives its own fair value entirely from D_1, K_e, and g — but noting the gap between the model output and the observed market price is exactly the kind of practical interpretation that turns a mechanical calculation into a complete answer.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly derives the required return via CAPM (9.6%), correctly calculates D1 (€1.575), correctly applies the Gordon Growth formula to reach ≈€34.24, and comments on how this compares to the given current market price.
5–7	5–7	Correct Gordon Growth application but either omits the CAPM step (e.g. mistakenly using the beta or risk-free rate directly as Ke) or uses D0 instead of D1 in the final formula.
2–4	2–4	Attempts CAPM or the Gordon Growth formula but with a fundamental setup error in one or both.
0–1	0–1	No attempt, or the €50 given price used as if it were an input needed by the model, rather than the output being compared to it.

Question 2 Total: 50 Marks

Question 3

Leverage, Breakeven Analysis & Capital Structure Costs (50 Marks)

Part (a)

10 Marks

Define and explain leverage, business risk, sales risk, operating risk, and financial risk.

FULL MODEL ANSWER

Leverage. The use of fixed costs (either fixed operating costs, like depreciation and salaried staff, or fixed financial costs, like interest on debt) to magnify the effect of a change in sales on a company's earnings. Leverage doesn't create risk from nothing — it amplifies whatever underlying variability already exists in sales, so a given percentage change in sales produces a larger percentage change in EBIT (operating leverage) or in earnings available to shareholders (financial leverage).

Business risk. The overall variability, or uncertainty, in a firm's operating income (EBIT) that arises from the nature of its business and operations — independent of how the firm is financed. Business risk combines sales risk and operating risk (below); a firm's underlying business risk is a function of its industry, competitive position, and cost structure, regardless of its capital structure choices.

Sales risk. The portion of business risk that comes from uncertainty in the firm's unit sales volume and the price obtained for those units — i.e. uncertainty about revenue itself, before any consideration of the firm's cost structure. A firm in a cyclical industry, or one exposed to volatile commodity prices, faces higher sales risk than one with stable, predictable demand and pricing.

Operating risk. The additional risk introduced by the firm's mix of fixed and variable operating costs, on top of whatever sales risk already exists. A firm with a high proportion of fixed operating costs relative to variable costs (high operating leverage) will see a given percentage change in sales translate into a larger percentage change in EBIT than a firm with mostly variable costs — even facing identical sales risk, the firm with more fixed operating costs bears more operating risk.

Financial risk. The additional risk borne by shareholders because the firm uses debt (a fixed financial cost, in the form of interest) in its capital structure. Financial risk is entirely a function of financing choices, not operations — a firm with zero debt has zero financial risk regardless of its business or operating risk, while adding debt increases the variability of earnings available to shareholders for any given level of EBIT variability, since interest must be paid regardless of how the business performs.

The key relationship to make explicit: **total risk to shareholders = business risk + financial risk**, and business risk itself decomposes into sales risk and operating risk. A clear answer shows how these concepts nest inside each other, rather than treating all five terms as unrelated definitions sitting side by side.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly and precisely defines all five terms, AND explicitly explains how they relate to each other (sales risk + operating risk = business risk; business risk + financial risk = total risk to shareholders).
5–7	5–7	Correctly defines most or all terms individually but without explaining how they relate to or build on each other.
2–4	2–4	Defines two or three of the five terms correctly, with the others vague or missing.
0–1	0–1	No attempt, or definitions that don't distinguish between the different types of risk at all.

Jessica is analysing the potential impact of an improving economy on earnings at MST plc, one of Ireland's largest medical device companies producing replacement hips. MST is headquartered in Dublin. This year the company has generated €93,000 in revenue. Jessica projects that sales will improve by 10% next year due to increased demand for replacement hips. She wants to see how MST's earnings might respond given that level of increase in sales.

MST sold 6,000 hips in 2023. The average price per hip was €24, fixed costs associated with production total €15,000 per year, and variable costs are €14 per hip.

A source-paper note before starting: the introduction above states MST "generated €93,000 in revenue" this year, but 6,000 hips × €24 per hip = €144,000, not €93,000 — the two figures don't reconcile. Since every calculation required in parts (i)–(vii) below is built directly and exclusively from the explicitly given 6,000 hips, €24 price, €15,000 fixed costs, and €14 variable cost per hip (and these produce clean, internally consistent answers throughout, cross-checked via the income statement in part (vii)), those figures are used as the basis for everything below. The €93,000 revenue figure and the 10% sales growth projection appear to be scene-setting context left over from an earlier draft of the question, rather than inputs the seven specific calculations actually require.

Part (b)(i) **2 Marks**

(i) What is the degree of operating leverage?

FULL MODEL ANSWER

Sales = 6,000 × €24 = €144,000
Variable costs = 6,000 × €14 = €84,000
Contribution margin = €144,000 − €84,000 = €60,000
EBIT = €60,000 − €15,000 (fixed costs) = €45,000
DOL = Contribution margin ÷ EBIT = €60,000 ÷ €45,000
DOL ≈ 1.33

Part (b)(ii)

2 Marks

(ii) MST also employs debt financing. If MST can borrow at 8%, the interest cost is €40,000. What is the degree of financial leverage of MST if 6,000 hips are produced and sold?

FULL MODEL ANSWER

EBT = EBIT − Interest = €45,000 − €40,000 = €5,000
DFL = EBIT ÷ EBT = €45,000 ÷ €5,000
DFL = 9.00

The €40,000 interest cost is given directly — note it does **not** equal 8% of any figure stated elsewhere in the question, so it should simply be taken at face value rather than recalculated from the borrowing rate.

Part (b)(iii)

2 Marks

(iii) Calculate MST's total leverage.

FULL MODEL ANSWER

DTL = DOL × DFL = 1.33 × 9.00
DTL = 12.00

Cross-check: DTL = Contribution margin ÷ EBT = €60,000 ÷ €5,000 = 12.00 — matches exactly, confirming the DOL and DFL figures above are correct.

Part (b)(iv)

2 Marks

(iv) Calculate the per unit contribution margin?

FULL MODEL ANSWER

CM per unit = Price per unit − Variable cost per unit
= €24 − €14
= €10 per hip

Part (b)(v) **2 Marks**

(v) Calculate the operating breakeven point for MST?

FULL MODEL ANSWER

Operating breakeven = Fixed costs ÷ CM per unit
= €15,000 ÷ €10
= 1,500 hips

This is the sales level at which EBIT = €0 — it considers only the fixed *operating* costs (€15,000), not the fixed financial cost of interest, which is why it's lower than the total breakeven point in part (vi).

Part (b)(vi) **2 Marks**

(vi) Calculate the total breakeven point?

FULL MODEL ANSWER

Total breakeven = (Fixed costs + Interest) ÷ CM per unit
= (€15,000 + €40,000) ÷ €10
= 5,500 hips

This is the sales level at which **net income** (EBT) = €0 — it adds the fixed financial cost of interest (€40,000) on top of the fixed operating costs, since shareholders only break even once both fixed operating costs AND the interest obligation are covered.

Part (b)(vii)

3 Marks

(vii) Verify your calculations in (iv) – (vi) by constructing an income statement for the breakeven sales.

FULL MODEL ANSWER

Building the income statement at the **total** breakeven point of 5,500 hips verifies all three preceding calculations in one pass — it should show net income of exactly zero:

	€
Sales (5,500 × €24)	132,000
Variable costs (5,500 × €14)	(77,000)
Contribution margin (5,500 × €10 CM/unit)	**55,000**
Fixed costs	(15,000)
EBIT	**40,000**
Interest	(40,000)
Net income (EBT)	**0**

Net income comes out at exactly €0, confirming the total breakeven point of 5,500 hips from part (vi) is correct. The contribution margin of €55,000 also reconciles to 5,500 × the €10 per-unit figure from part (iv), and if you additionally check the operating breakeven point of 1,500 hips from part (v), EBIT alone (before interest) comes out to exactly €0 at that lower sales level — all three answers tie together through this single check.

How the 15 Marks for Parts (b)(i)-(vii) Are Likely Allocated

Tier	Marks	What separates this tier
(i)-(iii) 6	6	2 marks each for DOL (1.33), DFL (9.00), and DTL (12.00), with own-figure credit if an earlier figure is carried through consistently.
(iv)-(vi) 6	6	2 marks each for the €10 per-unit contribution margin, the 1,500-hip operating breakeven, and the 5,500-hip total breakeven.
(vii) 3	3	A correctly constructed income statement at the total breakeven sales level (5,500 hips) that reconciles to exactly €0 net income; partial credit for building the statement at the operating breakeven (1,500 hips) instead, which correctly shows EBIT (not net income) at zero.

Part (c) **25 Marks**

Explain the costs of financial distress, the agency costs of equity, the costs of asymmetric information, as well as the potential effects of these costs on a company's optimal capital structure.

FULL MODEL ANSWER

Each of these three cost categories pulls a company's optimal capital structure in a different, and sometimes conflicting, direction.

1. Costs of Financial Distress

What it is. The costs a firm incurs as the probability of being unable to meet its debt obligations rises, split into direct costs (legal and administrative fees in a formal bankruptcy or restructuring) and, usually far larger, indirect costs (lost customers and suppliers demanding cash-on-delivery terms as they lose confidence in the firm's survival, key employees leaving for more stable employers, and managers becoming distracted by survival rather than value-creating decisions, sometimes leading to underinvestment in positive-NPV projects to conserve cash).

Effect on optimal capital structure. These costs rise with leverage, since more debt means a higher probability of being unable to meet fixed obligations in a downturn. This is the direct counterweight to the interest tax shield in trade-off theory: firms increase debt to capture the tax benefit only up to the point where the marginal expected cost of distress equals the marginal tax saving, producing an interior optimal leverage level rather than maximum possible debt.

2. Agency Costs of Equity

What it is. The costs arising from the separation of ownership (shareholders) and control (managers) in a firm financed heavily or entirely by outside equity. Managers holding only a small personal stake in the company bear only a fraction of the cost of value-destroying decisions — empire-building through unnecessary acquisitions, excessive perks, or overinvestment in low-return pet projects — while capturing most of the personal benefit, since the full cost is spread across all shareholders (Jensen and Meckling, 1976).

Effect on optimal capital structure. Debt can actually reduce this particular agency cost: the fixed obligation to make interest and principal payments forces managers to generate and disgorge cash rather than let it accumulate for discretionary (and potentially wasteful) spending — Jensen's (1986) free cash flow hypothesis. This pushes toward somewhat higher leverage than a pure financial-distress-cost analysis alone would suggest, since debt provides a genuine discipline benefit on top of the tax shield.

3. Costs of Asymmetric Information

What it is. The cost that arises because managers know more about the true value and prospects of the firm than outside investors do. When a firm issues new equity, investors can't fully distinguish whether it's because management genuinely needs the capital for good investment opportunities, or because management believes the shares are currently overvalued and wants to sell more of them while the price is favourable (Myers and Majluf, 1984). Because investors can't tell the difference, they rationally mark down the share price on any new equity issue announcement, making equity an expensive way to raise capital regardless of the company's actual underlying quality.

Effect on optimal capital structure. This is the foundation of the pecking order theory: because the information asymmetry problem is worst for equity and progressively less severe for debt (safe debt suffers from this problem the least, since its value is far less sensitive to what only insiders know about firm value), firms prefer to finance new investment with internal funds first, then debt, and only issue new equity as a last resort. Under this view, there's no actively targeted optimal leverage ratio at all — observed leverage is simply whatever falls out of firms following this financing hierarchy over time as their internal funds and investment needs happen to interact.

Bringing the three together

The three costs don't all point the same way. Financial distress costs argue for less debt as leverage rises; agency costs of equity argue debt has a genuine discipline benefit that pure tax-shield analysis misses; and information asymmetry costs argue firms should follow a financing hierarchy (internal funds, then debt, then equity) rather than target any specific ratio at all. A complete answer recognises that real firms' observed capital structures likely reflect some blend of all three effects simultaneously, rather than following any single theory in isolation — which is also why capital structure in practice varies so much across firms and industries facing different combinations of these three cost categories.

A strong answer doesn't just define the three cost categories in isolation — it explicitly traces each one through to its implication for the optimal amount of debt (too much distress cost argues down, agency cost of equity argues debt has discipline value, information asymmetry argues for a financing pecking order rather than a fixed target), and notes that these implications aren't all pulling in the same direction.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
19–25	19–25	Correctly explains all three cost categories with reference to the underlying literature (Jensen and Meckling, Jensen, Myers and Majluf), and explicitly connects each to its specific implication for optimal capital structure, showing that the three effects can pull in different directions.
13–18	13–18	Correctly explains two or three of the cost categories but with limited connection to capital structure implications, or without reference to the underlying literature.
7–12	7–12	General discussion of debt/equity trade-offs without specifically distinguishing financial distress costs, agency costs of equity, and information asymmetry costs as separate concepts.
0–6	0–6	Minimal or generic discussion with little real engagement with the three named cost categories.

Question 3 Total: 50 Marks